

1. Dirty Reads

Cu probleme:

```
--Dirty Reads
```

```
BEGIN TRANSACTION
```

```
UPDATE Client SET Varsta=60 where IDClient=1
```

```
INSERT INTO LogHistory(Place,Operation,Executed) VALUES('Client','Update',GETDATE())
```

```
WAITFOR DELAY '00:00:10'
```

```
ROLLBACK TRANSACTION
```

Transaction 1

```
--With Problem
```

```
SET TRANSACTION ISOLATION LEVEL READ UNCOMMITTED
```

```
BEGIN TRAN
```

```
SELECT * FROM Client
```

```
INSERT INTO LogHistory(Place,Operation,Executed) VALUES('Client','SELECT',GETDATE())
```

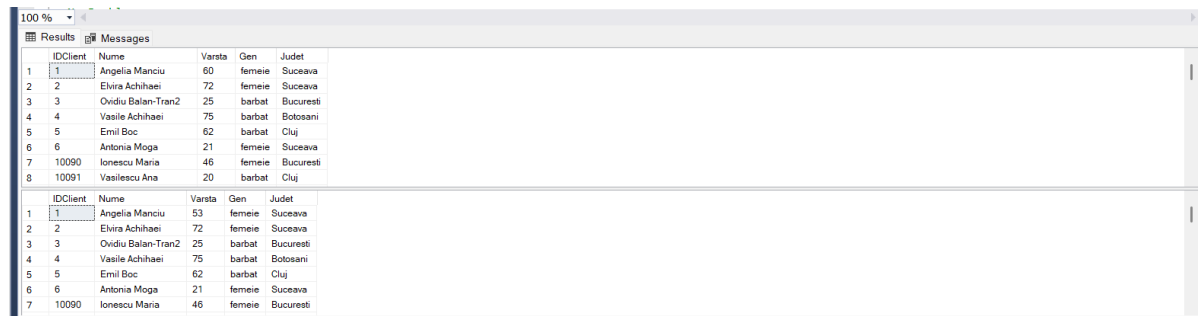
```
WAITFOR DELAY '00:00:15'
```

```
SELECT * FROM Client
```

```
INSERT INTO LogHistory(Place,Operation,Executed) VALUES('Client','SELECT',GETDATE())
```

```
COMMIT TRAN
```

Transaction 2



IDClient	Nume	Varsta	Gen	Judet
1	Angelia Manci	60	femeie	Suceava
2	Elvira Achihaei	72	femeie	Suceava
3	Ovidiu Balan-Tran2	25	barbat	Bucuresti
4	Vasile Achihaei	75	barbat	Botosani
5	Emil Boc	62	barbat	Cluj
6	Antonia Moga	21	femeie	Suceava
7	Ionescu Maria	46	femeie	Bucuresti
8	Vasilescu Ana	20	barbat	Cluj

IDClient	Nume	Varsta	Gen	Judet
1	Angelia Manci	53	femeie	Suceava
2	Elvira Achihaei	72	femeie	Suceava
3	Ovidiu Balan-Tran2	25	barbat	Bucuresti
4	Vasile Achihaei	75	barbat	Botosani
5	Emil Boc	62	barbat	Cluj
6	Antonia Moga	21	femeie	Suceava
7	Ionescu Maria	46	femeie	Bucuresti

Rezolvare:

--Modificam Transactia 2

```
BEGIN TRANSACTION
```

```
UPDATE Client SET Varsta=57 where IDClient=1
```

```
INSERT INTO LogHistory(Place,Operation,Executed) VALUES('Client','Update',GETDATE())
```

```
WAITFOR DELAY '00:00:10'
```

```
ROLLBACK TRANSACTION
```

Transaction 1

```
--No Problem
SET TRANSACTION ISOLATION LEVEL READ COMMITTED
BEGIN TRAN
SELECT * FROM Client
INSERT INTO LogHistory(Place,Operation,Executed) VALUES('Client','SELECT',GETDATE())
WAITFOR DELAY '00:00:15'
SELECT * FROM Client
INSERT INTO LogHistory(Place,Operation,Executed) VALUES('Client','SELECT',GETDATE())
COMMIT TRAN
```

Transaction 2

Results					
IDClient	Nume	Varsta	Gen	Judet	
1	Angelia Manciu	53	femeie	Suceava	
2	Elvira Achihai	72	femeie	Suceava	
3	Ovidiu Balan-Tran2	25	barbat	Bucuresti	
4	Vasile Achihai	75	barbat	Botosani	
5	Emil Boc	62	barbat	Cluj	
6	Antonia Moga	21	femeie	Suceava	
7	10090 Ionescu Maria	46	femeie	Bucuresti	
8	10091 Vasilescu Ana	20	barbat	Cluj	

2. Non-repeatable Reads

Cu probleme:

```
INSERT INTO Client(Nume,Varsta,Gen,Judet) VALUES('Leon XIV',67,'barbat','USA')
INSERT INTO LogHistory(Place,Operation,Executed) VALUES('Client','Insert',GETDATE())
BEGIN TRANSACTION
WAITFOR DELAY '00:00:10'
UPDATE Client SET Varsta=70 where Nume='Leon XIV'
INSERT INTO LogHistory(Place,Operation,Executed) VALUES('Client','Update',GETDATE())
COMMIT TRAN
```

Transaction 1

```
--With Problem
SET TRANSACTION ISOLATION LEVEL READ COMMITTED
BEGIN TRAN
SELECT * FROM Client
INSERT INTO LogHistory(Place,Operation,Executed) VALUES('Client','SELECT',GETDATE())
WAITFOR DELAY '00:00:15'
SELECT * FROM Client
INSERT INTO LogHistory(Place,Operation,Executed) VALUES('Client','SELECT',GETDATE())
COMMIT TRAN
```

Transaction 2

Results					
IDClient	Nume	Varsta	Gen	Judet	
112	10219 Mugurel Achihai	49	barbat	Suceava	
113	10220 Mugurel Manciu	49	barbat	Suceava	
114	10221 Mugurel Achihai	49	barbat	Suceava	
115	10222 Robert Prevost	70	barbat	USA	
116	10223 Robert Prevost	70	barbat	USA	
117	10224 Francisc	88	barbat	Argentina	
118	10225 Francisc	88	barbat	Argentina	
119	10226 Leon XIV	67	barbat	USA	

Rezolvare:

```

INSERT INTO Client(Nume,Varsta,Gen,Judet) VALUES('Benedict XVI',80,'barbat','Germania')
INSERT INTO LogHistory(Place,Operation,Executed) VALUES('Client','Insert',GETDATE())
BEGIN TRANSACTION
WAITFOR DELAY '00:00:10'
UPDATE Client SET Varsta=95 where Nume='Benedict XVI'
INSERT INTO LogHistory(Place,Operation,Executed) VALUES('Client','Update',GETDATE())
COMMIT TRAN

```

Transaction 1

```

--No Problem
SET TRANSACTION ISOLATION LEVEL REPEATABLE READ
BEGIN TRAN
SELECT * FROM Client
INSERT INTO LogHistory(Place,Operation,Executed) VALUES('Client','SELECT',GETDATE())
WAITFOR DELAY '00:00:15'
SELECT * FROM Client
INSERT INTO LogHistory(Place,Operation,Executed) VALUES('Client','SELECT',GETDATE())
COMMIT TRAN

```

Transaction 2

113	10220	Mugurel Manci	49	barbat	Suceava
114	10221	Mugurel Achihai	49	barbat	Suceava
115	10222	Robert Prevost	70	barbat	USA
116	10223	Robert Prevost	70	barbat	USA
117	10224	Francisc	88	barbat	Argentina
118	10225	Francisc	88	barbat	Argentina
119	10226	Leon XIV	70	barbat	USA
120	10227	Benedict XVI	80	barbat	Germa...

IDClient	Nume	Varsta	Gen	Judet
113	10220	Mugurel Manci	49	barbat
114	10221	Mugurel Achihai	49	barbat
115	10222	Robert Prevost	70	barbat
116	10223	Robert Prevost	70	barbat
117	10224	Francisc	88	barbat
118	10225	Francisc	88	barbat
119	10226	Leon XIV	70	barbat
120	10227	Benedict XVI	80	barbat

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Results Messages						
	IDClient	Nume	Varsta	Gen	Judet	
105	10198	Dumitrescu Elena	53	femeie	Bucuresti	
106	10199	Stanescu Andrei	48	barbat	Cluj	
107	10213	Mugurel Manci	54	barbat	Suceava	
108	10214	Mugurel Manci	54	barbat	Suceava	
109	10215	Mugurel Achihaei	49	barbat	Suceava	
110	10216	Mugurel Achihaei	49	barbat	Suceava	
111	10217	Mugurel Achihaei	49	barbat	Suceava	
112	10219	Mugurel Achihaei	49	barbat	Suceava	
113	10220	Mugurel Manci	49	barbat	Suceava	
114	10221	Mugurel Achihaei	49	barbat	Suceava	
115	10222	Robert Prevost	70	barbat	USA	
116	10223	Robert Prevost	70	barbat	USA	
117	10224	Francisc	88	barbat	Argentina	
118	10225	Francisc	88	barbat	Argentina	
119	10226	Leon XIV	70	barbat	USA	
120	10227	Benedict XVI	95	barbat	Germa...	

SELECT * FROM CLIENT

3. Phantom Reads

Cu probleme:

```
-----  
BEGIN TRAN  
WAITFOR DELAY '00:00:10'  
INSERT INTO Client(Nume,Varsta,Gen,Judet) VALUES('John Paul II',88,'barbat','Poland')  
INSERT INTO LogHistory(Place,Operation,Executed) VALUES('Client','Insert',GETDATE())  
COMMIT TRAN
```

Transaction 1

```
--With Problem  
SET TRAN ISOLATION LEVEL REPEATABLE READ  
BEGIN TRAN  
SELECT * FROM Client  
INSERT INTO LogHistory(Place,Operation,Executed) VALUES('Client','SELECT',GETDATE())  
WAITFOR DELAY '00:00:15'  
SELECT * FROM Client  
INSERT INTO LogHistory(Place,Operation,Executed) VALUES('Client','SELECT',GETDATE())  
COMMIT TRAN
```

Transaction 2

Results		Messages			
113	10220	Mugurel Manciu	49	barbat	Suceava
114	10221	Mugurel Achihaei	49	barbat	Suceava
115	10222	Robert Prevost	70	barbat	USA
116	10223	Robert Prevost	70	barbat	USA
117	10224	Francisc	88	barbat	Argentina
118	10225	Francisc	88	barbat	Argentina
119	10226	Leon XIV	70	barbat	USA
120	10227	Benedict XVI	95	barbat	Germa...

	IDClient	Nume	Varsta	Gen	Judet
114	10221	Mugurel Achihaei	49	barbat	Suceava
115	10222	Robert Prevost	70	barbat	USA
116	10223	Robert Prevost	70	barbat	USA
117	10224	Francisc	88	barbat	Argentina
118	10225	Francisc	88	barbat	Argentina
119	10226	Leon XIV	70	barbat	USA
120	10227	Benedict XVI	95	barbat	Germa...
121	10228	John Paul II	88	barbat	Poland

Solutie:

```
-----  
BEGIN TRAN  
WAITFOR DELAY '00:00:10'  
INSERT INTO Client(Nume,Varsta,Gen,Judet) VALUES('John Paul I',55,'barbat','Italia')  
INSERT INTO LogHistory(Place,Operation,Executed) VALUES('Client','Insert',GETDATE())  
COMMIT TRAN  
-----
```

Transaction 1

```
--No Problem
SET TRAN ISOLATION LEVEL SERIALIZABLE
BEGIN TRAN
SELECT * FROM Client
INSERT INTO LogHistory(Place,Operation,Executed) VALUES('Client','SELECT',GETDATE())
WAITFOR DELAY '00:00:15'
SELECT * FROM Client
INSERT INTO LogHistory(Place,Operation,Executed) VALUES('Client','SELECT',GETDATE())
COMMIT TRAN
```

Transaction 2

Results					
Messages					
114	10221	Mugurel Achihaei	49	barbat	Suceava
115	10222	Robert Prevost	70	barbat	USA
116	10223	Robert Prevost	70	barbat	USA
117	10224	Francisc	88	barbat	Argentina
118	10225	Francisc	88	barbat	Argentina
119	10226	Leon XIV	70	barbat	USA
120	10227	Benedict XVI	95	barbat	Germa...
121	10228	John Paul II	88	barbat	Poland

Results					
Messages					
	IDClient	Nume	Varsta	Gen	Judet
107	10213	Mugurel Manciu	54	barbat	Suceava
108	10214	Mugurel Manciu	54	barbat	Suceava
109	10215	Mugurel Achihaei	49	barbat	Suceava
110	10216	Mugurel Achihaei	49	barbat	Suceava
111	10217	Mugurel Achihaei	49	barbat	Suceava
112	10219	Mugurel Achihaei	49	barbat	Suceava
113	10220	Mugurel Manciu	49	barbat	Suceava
114	10221	Mugurel Achihaei	49	barbat	Suceava
115	10222	Robert Prevost	70	barbat	USA
116	10223	Robert Prevost	70	barbat	USA
117	10224	Francisc	88	barbat	Argentina
118	10225	Francisc	88	barbat	Argentina
119	10226	Leon XIV	70	barbat	USA
120	10227	Benedict XVI	95	barbat	Germa...
121	10228	John Paul II	88	barbat	Poland
122	10229	John Paul I	55	barbat	Italia

SELECT * FROM CLIENT